



Automated Classification of Peripheral Blood Cells Using an Efficient Convolutional Neural Network Architecture

Birmingham City University
School of Computing and Digital Technology

Coursework Report

Module Code: CMP6232
Module Title: Deep Neural Networks and Ethics
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Word Count: ~1500

August 1, 2026

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1 Introduction

Complete blood count (CBC) is one of the most commonly ordered tests in clinical medicine and is useful in the diagnosis of a variety of diseases, including anaemias, infections and hematological malignancies, like leukaemia. Manual determination of blood cell types by light microscopy is still used to confirm abnormal findings, but is time consuming, manual and subject to inter-observer error. The automatic classification of red blood cells (RBCs), white blood cells (WBCs), and platelets from microscopic images may speed up screening and make it more consistent, especially in high-volume or resource-limited environments.

This document outlines a proposal of the dataset to be studied and the data science problem to be attacked (Section 2), a survey of the previous related research work (Section 3), that should be used together as the background for suggesting the proposed efficient transfer-learning CNN model (Section 5).

2 Motivation and Objectives

This section explores the chosen dataset, the practical constraints of the project, and the data science problem being addressed.

2.1 Dataset

The main data source is **TXL-PBC** a carefully curated and re-annotated blood cell dataset that was created by integrating four publicly available resources namely Blood Cell Count Data (BCCD) BCDD PBC, and Raabin-WBC. It was released and a peer-reviewed data descriptor (Gan et al., 2025). Compared to a single dataset, such as using BCCD alone, TXL-PBC roughly multiplies the image count 3.5 times, resulting in 1,260 microscopy images and 18,143 bounding-box annotations that are mainly derived from images taken under varying staining and imaging conditions. TXLPBC comes with stratified sets of train/validation/test data that contain 882, 252, and 126 images respectively, and also with the bounding boxes of the same three classifications for which we will be generating single-cell patches from these images by the same procedure originally applied for a single-source dataset namely RBC, WBC, and Platelet.

Table 1: Structural breakdown and modelling implications of the derived dataset.

Property	Description	Implication for Modelling
Source Images	1,260 images pooled from BCCD, BCDD, PBC, and Raabin-WBC	The split is 882/252/126, so I don't need to do my own split.
Classes	RBC, WBC, Platelet (3 classes)	Standard 3-class problem. I'll use softmax with categorical crossentropy.
Class Imbalance	RBC: 89.9%, WBC: 7.2%, Platelet: 3.0% (18,143 annotations)	I'll compute class weights = $\frac{\text{total_samples}}{\text{n_classes} \times \text{class_count}}$. This is essential.
Variability	Images come from four different sources with different staining and scanners	I'll add colour normalisation and augmentation to make the model robust to lab differences.

2.2 Data Science Problem

The task is **multi- class image classification problem**: the aim is to train a neural network that is parameterized $f_{\theta}(x)$ such that, given a cropped image patch x , a right label $y \in \{\text{RBC, WBC, Platelet}\}$ is predicted. Besides getting a right label, this is also a **knowledge**

discovery problem: identifying the visual features learned by the network for each blood cell type, cell size, nuclear-to-cytoplasm ratio, staining intensity - the network relates to, so one knows what the cause of a misclassification is and can refer to the morphological criteria hematologists use.

This problem is affected by two technical challenges. For one thing, RBCs greatly exceed the numbers of WBCs and platelets in healthy blood samples, which creates an extreme class imbalance that demands loss-function weighting instead of just accuracy maximization. And, even though TXL-PBC is bigger and more diverse than a single-source blood cell dataset 1 260 microscopy images is still quite small in comparison with what modern CNNs can handle, so the focus during training should be on preventing overfitting. This way, the training should be carried out by taking advantage of a pretrained network via transfer learning, accompanied by data augmentation of the cropped cell images, rather than starting training from scratch.

3 Related Work

3.1 Traditional Machine Learning Methods

In the early days, automatic cell analysis still used traditional, manually chosen feature extractions methods. For example, [Ramoser et al. \(2005\)](#) used changes of image colourspace, morphology operations and edge detectors to separate the leukocyte cells, then used geometrical and textural descriptors as inputs to the classifiers like Support Vector Machines. Although these methods were computationally inexpensive, the pipeline was fragile: degradation of the segmentation happened with staining variation and uneven illumination, and the manually defined features generalised poorly across the changes of different laboratories and equipments.

3.2 Deep Neural Networks

3.2.1 Transfer Learning and Detection-Based Approaches

With the introduction of Convolutional Neural Networks (CNNs), the focus gradually changed to end-to-end representation learning. Transfer learning has been widely adopted in medical imaging because of its pre-training a network on big datasets like ImageNet, which allows the network to start with weights already learned ([Davamani et al., 2024](#)). That way, the pretrained filters already contain general-purpose information about image edges and textures. That should translate well to a microscopic domain. Apart from the models aimed First and foremost at classification, there are detection-oriented hybrid models combining the CNN layers and the transformer parts. One such example, CST-YOLO ([Kang et al., 2023](#)), was employed for blood cell detection, which was possible thanks to the active parallel development of blood cell databases and detection models that the data source of this proposal relies upon.

3.2.2 Efficient CNN Architectures

Three major CNN (Deep Convolution Neural Network) algorithms were chosen for this project, their summary in Tab. 2. Latest experimental studies show each of these approaches after being fine-tuned on blood cell and leukocyte datasets has yielded precision similar to much more complex ResNet-50 baselines. ([Davamani et al., 2024](#); [Ahmed, 2025](#)).

Table 2: Candidate CNN algorithms considered for this project.

Algorithm	Params	Key Mechanism	Trade-off	Chosen
EfficientNet-B0 (Tan and Le, 2019)	5.3M	Scales depth, width, and image size together for better efficiency	Best balance of accuracy and model size. Works well on medical images.	✓
MobileNetV3-Small (Howard et al., 2019)	2.5M	Uses lightweight depth-wise convolutions to keep the model small	Very fast and small, but might be too simple for this task.	–
DenseNet121 (Huang et al., 2017)	8.0M	Reuses features from earlier layers to improve accuracy	Good accuracy, but needs more memory and takes longer to train.	–

4 Concepts of Deep Learning

4.1 Transfer Learning and Convolutional Feature Extraction

Transfer learning uses the convolutional layers of a network trained on a large source dataset (usually ImageNet) as a fixed or slightly-tuned feature extractor for a new smaller target dataset. Since the network doesn't have to learn to detect edges, texture, and shape, it just learns to recombine previously-useful features for the task at hand. This is usually implemented by removing the original 1000-class ImageNet output layer, passing the final convolutional feature maps through global average pooling to turn the (spatial?) dimensions into a small feature vector, then adding a small, target class trained head.

4.2 Efficient Architectural Mechanisms

As shown in Table 2, different mechanisms that lead algorithms to efficiency are: compound scaling of depth, width, and resolution (EfficientNet); depthwise separable convolutions with squeeze-and-excitation recalibration (MobileNetV3); and dense feature reuse across layers (DenseNet). While these mechanisms mainly involve swapping capacity, parameter count, and memory footprint, they also lead to different choices, which is why an empirical comparison in Section 5 would be quite useful.

5 Proposed Model

5.1 Architecture Choice

From the three CNN architectures listed in Table 2, **EfficientNet-B0** is selected for this project. Even though MobileNetV3-Small is the smallest model, its limited power might render it unable to detect slight variations in blood cell types. While the DenseNet121 model has a high performance, it is too large and uses up too much memory for this three-class classification problem. To balance the accuracy and efficiency, EfficientNet-B0 uses compound scaling, which gives a better efficiency vs accuracy trade-off than EfficientNet-B2 ([Tan and Le, 2019](#)). It has also performed competitively on comparable classification tasks on blood cells to larger ResNet-50 models ([Davamani et al., 2024](#); [Ahmed, 2025](#)) which makes it a suitable model to use for this dataset.

The main part of pretrained convolutions consists in the extraction of features, while the ImageNet classification layer is substituted by the classification head that integrates global average pooling, dropout, and a three-unit softmax output layer for RBC, WBC, and Platelet classification. The Figure 1 represents the workflow.

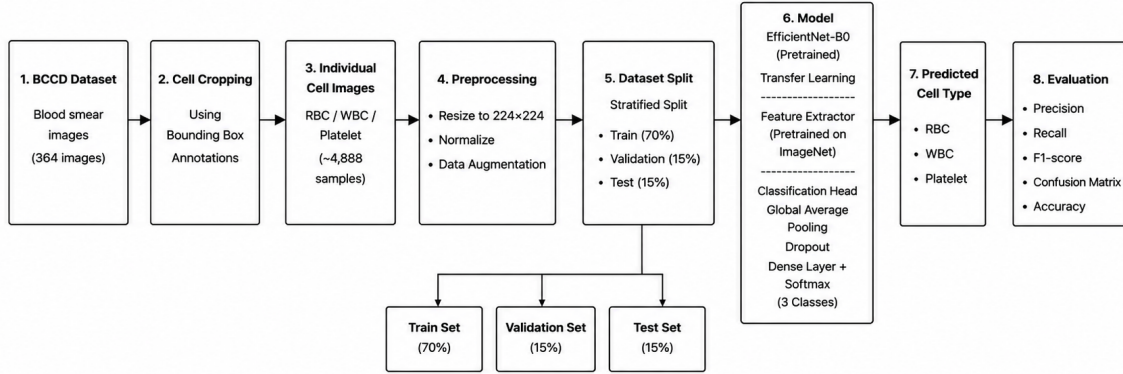


Figure 1: The complete workflow for the proposed blood cell classification framework.

5.2 Preparing the Dataset

5.2.1 Data Splitting Strategy

This partition of 882, 252, and 126 images in the TXL-PBC training, validation, and testing set, respectively, will remain as is for the resultant cell cropping dataset, since this will provide consistency with the format used by the original images, as well as enable future benchmarking.

5.2.2 Preprocessing

Cropped cell images will be scaled up to dimensions 224×224 and then standardised as the ImageNet dataset's mean values and standard deviations. With an aim to balance data and prevent overfitting, we will be using data augmentation strategies including random rotation, horizontal flipping, and subtle changes of brightness/contrast on the training part of the dataset, Mostly the classes WBC and Platelet, which are underrepresented.

5.3 Training the Model

During model training, class-weighted categorical cross-entropy loss together with Adam optimizer is employed. But, due to the existence of such imbalanced classes as stated in Section 2, the performance indicators will include macro-averaged of Precision, Recall, and F1-score instead of accuracy, while a normalized confusion matrix supports the outcome. In addition, training time and memory requirement are also monitored for the models so that different computational efficiencies can be recognized.

6 Conclusion

The current proposal suggests using the transfer learning technique to classify RBCs, WBCs, and Platelets with the help of TXL-PBC dataset. A detailed investigation of state-of-the-art classification methods reveals that EfficientNet-B0 is the optimal choice for the problem at hand because of its high accuracy, computational efficiency, and applicability to the dataset (Tan and Le, 2019; Davamani et al., 2024; Ahmed, 2025). The proposed methodology involves

cell isolation, data preprocessing, transfer learning via fine-tuning a previously trained model, and evaluation using Precision Recall F1-score, confusion matrix (macro-average)). The next phase of the project is to actually implement the method that has been conceived, and test it on TXL-PBC data.

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